Application of Machine Learning Algorithms to Predict Hiring Outcomes

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Background

Hiring Decision Dataset

The dataset, shared by Mr. Rabie El Kharoua 4 months ago in Kaggle, consists of 1500 datas from contestants and their factors influencing the hiring decision.

Source :
https://www.kaggle.com/datasets/rabieelkharoua/predicting-hiring-decisions-in-recruitment-data/data

Introductions

Dataset Preview

0 : Male
1 : Female

0: Not hired
1: Hired

	Age	Gender	Education Level	Experience Years	Previous Companies	Distance From Company	Interview Score	Skill Score	Personality Score	Recruitment Strategy	Hiring Decision
0	26	1	2	0	3	26.783828	48	78	91	1	1
1	39	1	4	12	3	25.862694	35	68	80	2	1
2	48	0	2	3	2	9.920805	20	67	13	2	0
3	34	1	2	5	2	6.407751	36	27	70	3	0
4	30	0	1	6	1	43.105343	23	52	85	2	0

1: Bachelor's (Type 1)
2: Bachelor's (Type 2)
3: Master's
4: PhD

1: Aggressive
2: Moderate
3: Conservative

Modules

Jupyter Notebook

Numpy

Pandas

Scikit-Learn

Matplotlib

K - Nearest Neighbor

Logistic Regression

Naive Bayes

Decision Trees

LightGBM

Support Vector Machine

Random Forests

Gradient Boosting

Ada Boost

XG Boost

Methodology

```
Age                      0
Gender                   0
Education Level          0
Experience Years         0
Previous Companies       0
Distance From Company    0
Interview Score          0
Skill Score              0
Personality Score        0
Recruitment Strategy     0
Hiring Decision          0
dtype: int64
```

Check for missing data : no data is missing.

1

2

3

```
Fold 1:
Training data: (1350, 11), Hired (1): 427, Not Hired (0): 923
Test data: (150, 11)
Fold 2:
Training data: (1350, 11), Hired (1): 418, Not Hired (0): 932
Test data: (150, 11)
Fold 3:
Training data: (1350, 11), Hired (1): 421, Not Hired (0): 929
Test data: (150, 11)
```

Cross-Validation: Using 10-cross validation.

1500 Datas → 1350 Training set → 437 Hired / 913 Not hired
1500 Datas → 150 Test set

1500 Datas → 1350 Training set → 913 Hired / 913 Not hired
1500 Datas → 150 Test set

Oversampling using SMOTE

Processing Data

Example of Results for each cross validation with Random Forests

Fold 1:
Classification Report:

	precision	recall	f1-score	support
0	0.96	0.96	0.96	112
1	0.87	0.87	0.87	38
accuracy			0.93	150
macro avg	0.91	0.91	0.91	150
weighted avg	0.93	0.93	0.93	150

FOLD 1

Fold 2:
Classification Report:

	precision	recall	f1-score	support
0	0.94	0.94	0.94	103
1	0.87	0.87	0.87	47
accuracy			0.92	150
macro avg	0.91	0.91	0.91	150
weighted avg	0.92	0.92	0.92	150

FOLD 2

Fold 3:
Classification Report:

	precision	recall	f1-score	support
0	0.89	0.94	0.92	106
1	0.84	0.73	0.78	44
accuracy			0.88	150
macro avg	0.87	0.84	0.85	150
weighted avg	0.88	0.88	0.88	150

FOLD 3

Fold 4:
Classification Report:

	precision	recall	f1-score	support
0	0.93	0.96	0.95	101
1	0.91	0.86	0.88	49
accuracy			0.93	150
macro avg	0.92	0.91	0.92	150
weighted avg	0.93	0.93	0.93	150

FOLD 4

Fold 5:
Classification Report:

	precision	recall	f1-score	support
0	0.97	0.94	0.95	100
1	0.89	0.94	0.91	50
accuracy			0.94	150
macro avg	0.93	0.94	0.93	150
weighted avg	0.94	0.94	0.94	150

FOLD 5

Fold 6:
Classification Report:

	precision	recall	f1-score	support
0	0.97	0.99	0.98	110
1	0.97	0.93	0.95	40
accuracy			0.97	150
macro avg	0.97	0.96	0.97	150
weighted avg	0.97	0.97	0.97	150

FOLD 6

Fold 7:
Classification Report:

	precision	recall	f1-score	support
0	0.94	0.98	0.96	108
1	0.95	0.83	0.89	42
accuracy			0.94	150
macro avg	0.94	0.91	0.92	150
weighted avg	0.94	0.94	0.94	150

FOLD 7

Fold 8:
Classification Report:

	precision	recall	f1-score	support
0	0.94	0.96	0.95	101
1	0.91	0.88	0.90	49
accuracy			0.93	150
macro avg	0.93	0.92	0.92	150
weighted avg	0.93	0.93	0.93	150

FOLD 8

Fold 9:
Classification Report:

	precision	recall	f1-score	support
0	0.92	0.99	0.95	100
1	0.98	0.82	0.89	50
accuracy			0.93	150
macro avg	0.95	0.91	0.92	150
weighted avg	0.94	0.93	0.93	150

FOLD 9

Fold 10:
Classification Report:

	precision	recall	f1-score	support
0	0.88	0.96	0.92	94
1	0.92	0.79	0.85	56
accuracy			0.89	150
macro avg	0.90	0.87	0.88	150
weighted avg	0.90	0.89	0.89	150

FOLD 10

Processing Data

Example of average classification for Random Forests

```
              precision    recall  f1-score     support
0              0.934376  0.962113  0.947772  103.500000
1              0.911008  0.850678  0.878617   46.500000
accuracy       0.927333  0.927333  0.927333    0.927333
macro avg      0.922692  0.906395  0.913194  150.000000
weighted avg   0.927747  0.927333  0.926504  150.000000
Average ROC AUC for Random Forest: 0.932
Average Confusion Matrix for Random Forest:
[[99.6  3.9]
 [ 7.  39.5]]
```

Processing Data

Average Classification Results of all Models

```
              precision    recall  f1-score     support
0              0.894284  0.820397  0.855464  103.500000
1              0.661432  0.785706  0.716995   46.500000
accuracy       0.809333  0.809333  0.809333    0.809333
macro avg      0.777858  0.803051  0.786229  150.000000
weighted avg   0.824078  0.809333  0.813494  150.000000
Average ROC AUC for KNN: 0.861
Average Confusion Matrix for KNN:
[[84.9 18.6]
 [10.  36.5]]
```

K - Nearest Neighbor

```
              precision    recall  f1-score  support
0              0.909719  0.820463  0.862320   103.50
1              0.672194  0.822561  0.737935    46.50
accuracy       0.820000  0.820000  0.820000     0.82
macro avg      0.790956  0.821512  0.800128   150.00
weighted avg   0.838144  0.820000  0.824423   150.00
Average ROC AUC for Logistic Regression: 0.887
Average Confusion Matrix for Logistic Regression:
[[84.9 18.6]
 [ 8.4 38.1]]
```

Logistic Regression

```
              precision    recall  f1-score     support
0              0.936155  0.965034  0.950084  103.500000
1              0.919859  0.854784  0.884723   46.500000
accuracy       0.930667  0.930667  0.930667    0.930667
macro avg      0.928007  0.909909  0.917403  150.000000
weighted avg   0.931588  0.930667  0.929952  150.000000
Average ROC AUC for LightGBM: 0.933
Average Confusion Matrix for LightGBM:
[[99.9  3.6]
 [ 6.8 39.7]]
```

LightGBM

```
              precision    recall  f1-score     support
0              0.914434  0.817398  0.862786  103.500000
1              0.670948  0.834637  0.742063   46.500000
accuracy       0.821333  0.821333  0.821333    0.821333
macro avg      0.792691  0.826018  0.802424  150.000000
weighted avg   0.840929  0.821333  0.825865  150.000000
Average ROC AUC for Naive Bayes: 0.890
Average Confusion Matrix for Naive Bayes:
[[84.6 18.9]
 [ 7.9 38.6]]
```

Naive Bayes

```
              precision    recall  f1-score     support
0              0.914911  0.895725  0.904972  103.500000
1              0.777382  0.815297  0.794828   46.500000
accuracy       0.870667  0.870667  0.870667    0.870667
macro avg      0.846147  0.855511  0.849900  150.000000
weighted avg   0.873787  0.870667  0.871545  150.000000
Average ROC AUC for Decision Tree: 0.856
Average Confusion Matrix for Decision Tree:
[[92.7 10.8]
 [ 8.6 37.9]]
```

Decision Trees

```
              precision    recall  f1-score     support
0              0.941072  0.961979  0.951290  103.500000
1              0.913107  0.867612  0.889267   46.500000
accuracy       0.932667  0.932667  0.932667    0.932667
macro avg      0.927090  0.914795  0.920278  150.000000
weighted avg   0.932672  0.932667  0.932163  150.000000
Average ROC AUC for XGBoost: 0.934
Average Confusion Matrix for XGBoost:
[[99.6  3.9]
 [ 6.2 40.3]]
```

XG Boost

Processing Data

Average Classification Results of all Models

Support Vector Machine

	precision	recall	f1-score	support
0	0.910055	0.895453	0.902241	103.500000
1	0.774102	0.806378	0.788209	46.500000
accuracy	0.866667	0.866667	0.866667	0.866667
macro avg	0.842078	0.850915	0.845225	150.000000
weighted avg	0.869362	0.866667	0.867036	150.000000

Average ROC AUC for SVM: 0.907
Average Confusion Matrix for SVM:

```
[[92.7 10.8]
 [ 9.2 37.3]]
```

Gradient Boosting

	precision	recall	f1-score	support
0	0.935293	0.940838	0.937896	103.500000
1	0.866974	0.857904	0.861578	46.500000
accuracy	0.914667	0.914667	0.914667	0.914667
macro avg	0.901133	0.899371	0.899737	150.000000
weighted avg	0.915047	0.914667	0.914487	150.000000

Average ROC AUC for Gradient Boosting: 0.933
Average Confusion Matrix for Gradient Boosting:

```
[[97.4  6.1]
 [ 6.7 39.8]]
```

Random Forests

	precision	recall	f1-score	support
0	0.934376	0.962113	0.947772	103.500000
1	0.911008	0.850678	0.878617	46.500000
accuracy	0.927333	0.927333	0.927333	0.927333
macro avg	0.922692	0.906395	0.913194	150.000000
weighted avg	0.927747	0.927333	0.926504	150.000000

Average ROC AUC for Random Forest: 0.932
Average Confusion Matrix for Random Forest:

```
[[99.6  3.9]
 [ 7.  39.5]]
```

Ada Boost

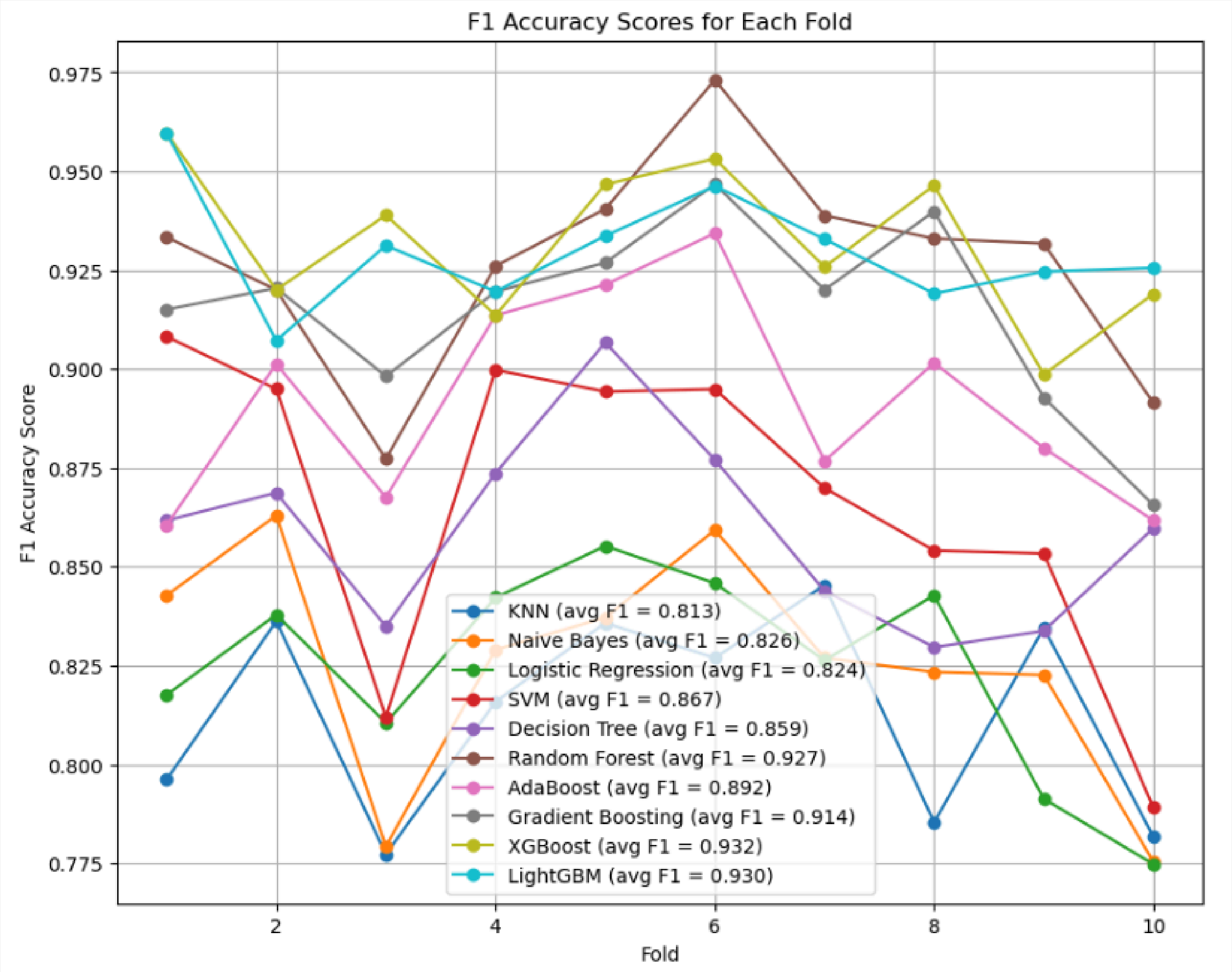
	precision	recall	f1-score	support
0	0.946750	0.890844	0.917486	103.50
1	0.786350	0.888485	0.832495	46.50
accuracy	0.890000	0.890000	0.890000	0.89
macro avg	0.866550	0.889664	0.874991	150.00
weighted avg	0.898323	0.890000	0.891833	150.00

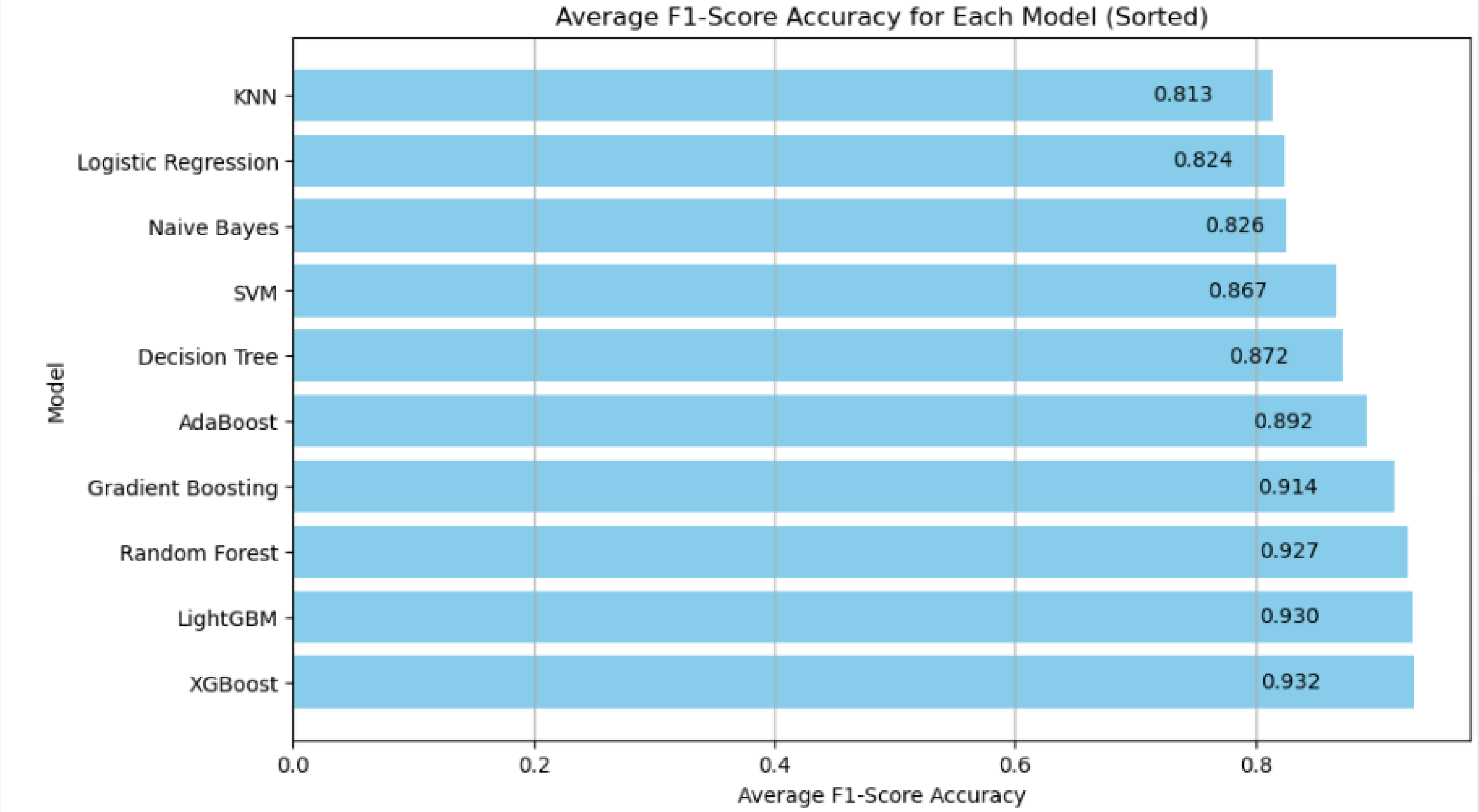
Average ROC AUC for AdaBoost: 0.928
Average Confusion Matrix for AdaBoost:

```
[[92.2 11.3]
 [ 5.2 41.3]]
```

Processing Data

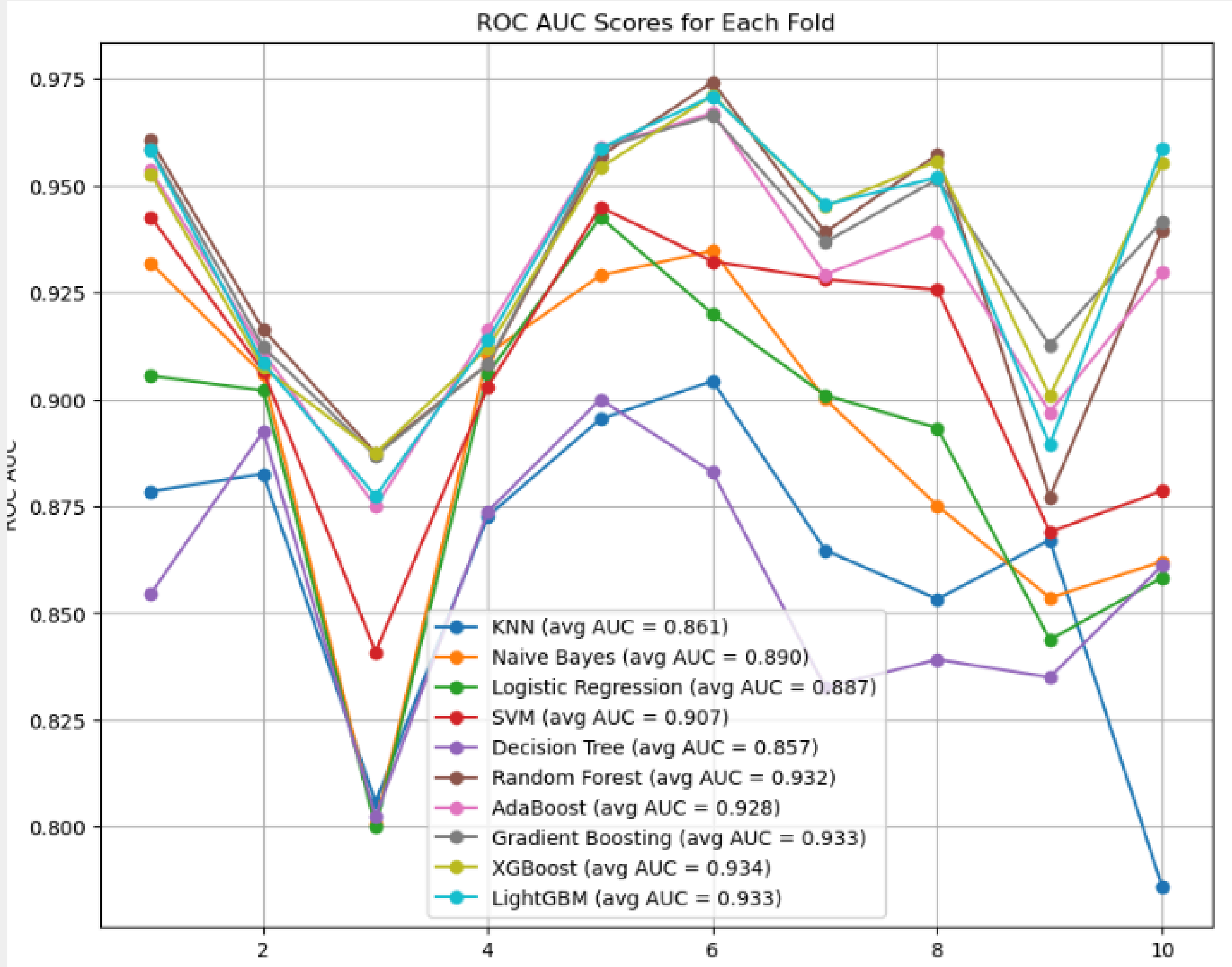
Visualizing Each Models F1 Scores

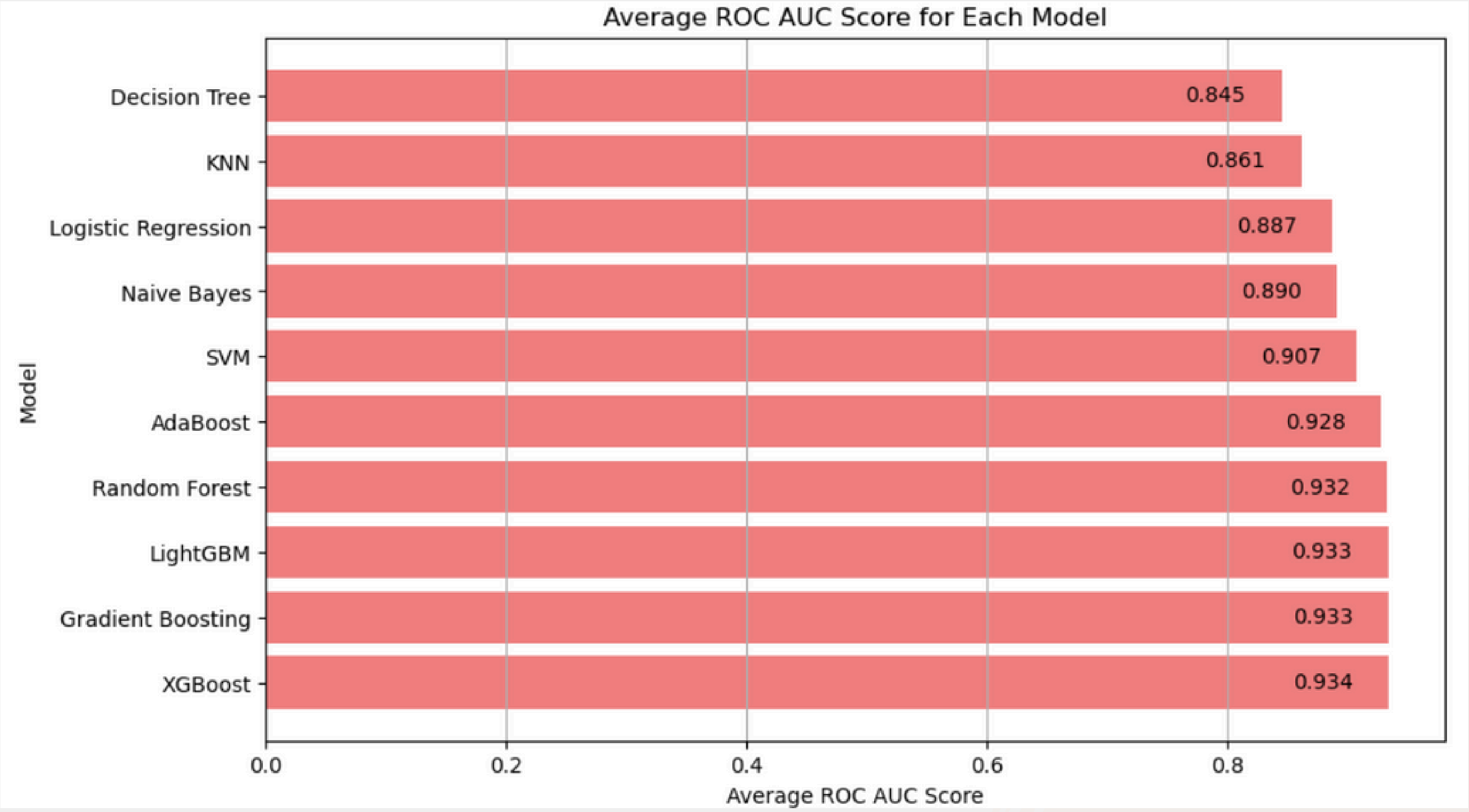




Processing Data

Visualizing Each Models ROC AUC Scores





Processing Data

Highest ROC AUC and Highest F1 Scores

ROC AUC Score

1. XG Boost
2. Gradient Boost
3. LightGBM

F1 Score

1. XG Boost
2. LightGBM
3. Random Forest

Processing Data

Feature Importance of the Top Scoring Models

This difference indicates that LightGBM is more sensitive to candidate-specific metrics than organizational factors. The results suggest that both organizational insights (like recruitment strategy) and candidate-specific evaluations (such as skill and interview scores) are significant in the hiring process.

Random Forests

Feature	Importance
Recruitment Strategy	0.351678
Education Level	0.110156
Skill Score	0.109007
Personality Score	0.101666
Interview Score	0.101431
Experience Years	0.083306
Distance From Company	0.057529
Age	0.045542
Previous Companies	0.029946
Gender	0.009740

Gradient Boosting

Feature	Importance
Recruitment Strategy	0.491521
Education Level	0.136432
Skill Score	0.095326
Personality Score	0.089477
Experience Years	0.077061
Interview Score	0.076281
Distance From Company	0.015817
Previous Companies	0.009147
Age	0.007122
Gender	0.001816

XG Boost

Feature	Importance
Recruitment Strategy	0.570039
Education Level	0.111624
Experience Years	0.063979
Interview Score	0.062636
Personality Score	0.059418
Skill Score	0.049440
Distance From Company	0.021608
Gender	0.021188
Previous Companies	0.020173
Age	0.019894

LightGBM

Feature	Importance
Interview Score	479.7
Skill Score	470.5
Personality Score	467.2
Experience Years	365.5
Distance From Company	339.3
Age	260.2
Education Level	255.1
Recruitment Strategy	164.9
Previous Companies	152.3
Gender	45.3

Conclusion

The feature importance analysis across the top-performing models—XGBoost, Random Forest, Gradient Boosting, and LightGBM—reveals that Recruitment Strategy is consistently the most influential factor in predicting hiring decisions for three of the models (XGBoost, Random Forest, and Gradient Boosting). This suggests that the recruitment approach used by the organization plays a pivotal role in determining hiring outcomes. Future works will focus on increasing more accuracy of the models.

Thank You So Much